

DMart Sales & Operational Intelligence Dashboard using Power BI

1. Abstract

The DMart Sales & Operational Intelligence Dashboard is an interactive business intelligence solution developed using Power BI to analyze e-commerce sales, customer behavior, operational efficiency, and revenue performance. The dashboard provides real-time insights into sales trends, customer demographics, payment methods, shipping efficiency, and product performance using advanced visualizations and DAX calculations.

The project aims to help business stakeholders make data-driven decisions by converting raw transactional data into meaningful visual insights. Advanced Power BI features such as KPI cards, slicers, drill-through analysis, heatmaps, funnel charts, scatter plots, bookmarks, and interactive filtering were implemented to enhance analytical capabilities and dashboard usability.

2. Introduction

Business organizations generate large volumes of transactional data every day. Analyzing this data manually becomes difficult and time-consuming. Business Intelligence tools such as Power BI help organizations transform raw data into actionable insights through dashboards and reports.

This project focuses on developing a professional Power BI dashboard for DMart ecommerce sales analysis. The dashboard provides insights related to revenue generation, customer behavior, product performance, shipping operations, marketing performance, and order cancellations.

3. Objectives

- To analyze e-commerce sales performance using Power BI.
- To monitor KPIs such as Total Revenue, Total Orders, and Customer Ratings.
- To identify top-performing products and states.
- To analyze customer demographics and engagement behavior.
- To evaluate shipping efficiency and cancellation rates.
- To create interactive dashboards using advanced Power BI visualizations.
- To enable data-driven business decision-making.

4. Tools and Technologies Used

Technology	Purpose
Power BI	Dashboard Development
DAX	Calculated Measures and KPIs
Power Query	Data Cleaning and Transformation
Excel/CSV Dataset	Data Source
Data Visualization	Business Insights

5. Dataset Description

The dataset contains e-commerce transactional data from DMart operations.

Main Attributes:

- Order ID
- Customer ID
- Product Name
- Category
- State
- Payment Method
- Order Status
- Shipping Charges
- Rating
- Discount Price
- Total Order Value
- Delivery Date
- Order Date

6. Data Preprocessing

The following preprocessing steps were performed:

- Handling missing values
- Converting date columns into the Date datatype
- Cleaning revenue fields
- Creating calculated columns
- Generating DAX measures
- Sorting month names using month numbers

7. Calculated Columns

1. **Delivery Days:** Used to calculate delivery duration between order and delivery date.
2. **Age Group:** Created for customer segmentation analysis.

3. **Month Name and Month Number:** Used for proper monthly trend analysis.

4. **DAX Measures**

The following measures were created:

- Total Revenue
- Total Orders
- Total Customers
- Average Rating
- Cancellation Rate
- Average Delivery Time
- Total Shipping Charges
- AOV (Average Order Value)

8. Dashboard Features

1. **Executive Dashboard:** Displays high-level KPIs and sales trends.

2. **Sales Analysis:** Analyzes revenue by category, state, and products.

3. **Customer Analysis:** Provides customer demographic and engagement insights.

4. **Operational Analysis:** Tracks shipping performance, cancellations, and payment status.

5. **Interactive Features**

- Slicers
- Bookmarks
- Drill-through
- Tooltips
- Smart Narrative
- Q&A Visual

6. **Visualizations Used**

- KPI Cards
- Line Chart
- Bar Chart
- Donut Chart
- Pie Chart
- Scatter Plot
- Funnel Chart
- Treemap
- Heatmap
- Stacked Column Chart

7. **Key Insights**

- Maharashtra generated the highest revenue.
- Local category products contributed maximum sales.
- Debit and UPI were the most used payment methods.
- Customers aged 21–40 formed the major customer base.
- Free shipping mode had higher order volume.
- Instagram generated significant marketing revenue.

- Cancellation rates were comparatively low.

9. Conclusion

The DMart Sales & Operational Intelligence Dashboard successfully transformed raw e-commerce data into actionable business insights using Power BI. The dashboard enables efficient monitoring of business KPIs, customer behavior, operational performance, and sales trends.

The project demonstrates practical implementation of business intelligence concepts, data visualization techniques, DAX calculations, and interactive dashboard design for real-world ecommerce analytics.

10. Future Enhancements

- Integration with real-time databases
- AI-based sales forecasting
- Customer churn prediction
- Mobile dashboard optimization
- Integration with cloud platforms